

ARTICLE

Gelatin interpenetration in poly *N*-isopropylacrylamide network reduces the compressive modulus of the scaffold: A property employed to mimic hepatic matrix stiffness

Joyita Sarkar¹  | Swapnil C. Kamble² | Rajendra Patil³ | Ashok Kumar^{4,5,6}  | Suresh W. Gosavi¹ 

¹Department of Physics, Savitribai Phule Pune University, Pune, Maharashtra, India

²Department of Technology, Savitribai Phule Pune University, Pune, Maharashtra, India

³Department of Biotechnology, Savitribai Phule Pune University, Pune, Maharashtra, India

⁴Department of Biological Sciences and Bioengineering, Indian Institute of Technology Kanpur, Kanpur, Uttar Pradesh, India

⁵Centre for Nanosciences, Indian Institute of Technology Kanpur, Kanpur, Uttar Pradesh, India

⁶Centre for Environmental Sciences and Engineering, Indian Institute of Technology Kanpur, Kanpur, Uttar Pradesh, India

Correspondence

Joyita Sarkar and Suresh W. Gosavi,
Department of Physics, Savitribai Phule Pune University, Ganeshkhind, Pune, Maharashtra 411007, India.
Email: joyita.scc12@gmail.com (J. S.); swg@physics.unipune.ac.in (S. W. G.)

Ashok Kumar, Department of Biological Sciences and Bioengineering, Indian Institute of Technology Kanpur, Kanpur, Uttar Pradesh 208016, India.
Email: ashokkum@iitk.ac.in (A. K.)

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Abstract

The progression of liver disease from normal to cirrhotic state is characterized by modulation of the stiffness of the extracellular matrix (ECM). Mimicking this modulation in vitro scaffold could provide a better insight into hepatic cell behavior. In this study, interpenetrating poly(*N*-isopropylacrylamide-co-gelatin) cryogels were synthesized in 48 different compositions to yield scaffolds of different properties. It was observed that a high concentration of *N*-isopropylacrylamide (NIPAAm) leads to the formation of small pores while gelatin interpenetration on poly-NIPAAm framework renders porous structure. Swelling properties and porosity of the gels decreased with an increase in NIPAAm concentration owing to the increased compactness of the gels. Gelatin interpenetration relaxed the gels and enhanced these properties. An increase in gelatin concentration led to a reduction in compressive moduli indicating that gelatin interpenetration in the poly-NIPAAm network softens the cryogel. With the increase in NIPAAm concentration, the effect of gelatin interpenetration in reducing the compressive moduli expanded. The cytocompatibility studies indicated that the gels are cell-adherent and compatible with HepG2. Furthermore, biochemical and real-time polymerase chain reaction studies revealed that HepG2 and Huh-7 cells cultured on scaffolds mimicking the ECM stiffness of normal liver (1.5–2.5 kPa) exhibited optimum liver-specific functionalities. Increasing the stiffness to fibrotic (4–9 kPa) and cirrhotic (10–20 kPa) ECM decreases the functionality.

KEYWORDS

compressive moduli, cryogel, liver extracellular matrix stiffness, polymer interpenetration

1 | INTRODUCTION

The stiffness of extracellular matrix (ECM) affects many cellular properties including proliferation, differentiation, morphogenesis and so forth (D'Amore et al., 2018; Saneyasu et al., 2016; Stewart et al., 2018; Zustiak et al., 2016). In the case of the liver, an

increase in stiffness of ECM as a result of extensive matrix deposition can be indicative of chronic disease. Techniques like noninvasive elastography and rheometry of the liver as a diagnostic tool rely on the change in stiffness of liver tissue upon disease progression (Wells, 2008). Mimicking the hepatic matrix stiffness in vitro could provide a broader insight into the cellular

processes as well as the progression of liver disease from normal to fibrotic and/or cirrhotic (Mueller & Sandrin, 2010). Conventional two-dimensional (2D) culture systems leads to dedifferentiation of hepatocytes and loss of hepatic function owing to high stiffness (modulus greater than 1 GPa) of 2D tissue culture plates (Cozzolino et al., 2016). 3D cell culture scaffolds have been extensively employed to mimic ECM properties. Gel-based scaffolds offer advantages over others owing to their tunable and easily-tailored properties (Geckil et al., 2010). The mechanical properties (tensile and compressive moduli, stiffness, swelling, etc.) of these scaffolds can be modulated by exploiting the polymer composition, concentration, and crosslinking (Wenger et al., 2007). Gelatin-methacryloyl hydrogels with tunable stiffness have been synthesized by exploiting the prepolymer concentration and ultraviolet (UV) exposure time; matrix stiffness regulated cytoskeleton-associated gene expression, cellular morphology and signaling pathways (Sun et al., 2019). On another occasion, an increase in modulus of collagen scaffolds by varied crosslinking enabled polarization of macrophages to an M2-like phenotype (Sridharan et al., 2019). In many occasions, combinations of synthetic and natural polymers have been employed to obtain desired and optimum mechanical properties of gel-based scaffolds. Using such a combination allows obtaining biocompatibility and proffer tunability to other physicochemical properties (Jain et al., 2009). One such instance is utilization of interpenetrating polymer networks that have been widely used to improve the mechanical strength of scaffolds for various tissue engineering applications, for example skin, bone, cartilage, and so forth (Chung et al., 2006; Gsib et al., 2017; Rennerfeldt et al., 2013; Van Vlierberghe et al., 2011). The stiffness of normal liver ranges from approximately 0.6 to 2 kPa, and could increase to approximately 20 kPa in a diseased state like cirrhosis (Yeh et al., 2002). Hepatocytes exhibit an arrest in proliferation and maintain differentiated phenotype when cultured on soft matrices in contrast to stiffer substrates where they are proliferative and dedifferentiated (Wells, 2008). Primary rat hepatocytes cultured on softer heparin gels of approximately 11 kPa revealed better liver-specific functionalities than those cultured on gels with approximately 27 and 116 kPa (You et al., 2013). Huh7.5 formed larger spheroids with better functionalities in a shorter time on polyethylene glycol-based hydrogels up to 2 kPa shear modulus indicating that tuning the substrate stiffness could be implemented for smart scaffold design (Lee et al., 2015). In addition, it was observed that resident liver stem cells undergo rapid differentiation to hepatocytes in substrates of 0.4 kPa (Cozzolino et al., 2016).

It can be concluded that establishing the relationship between stiffness of in vitro cell culture scaffold and cell functionality could bridge the gap between in vivo physiological conditions of cells and tissues. This could assist in the study of the progression and pathophysiology of liver diseases. Improving the functionality of hepatic cells in vitro could provide better performance in bioartificial liver devices as well as pharmaceutical drug screening platforms. As a 3D construct, cryogels have previously been

shown to be compatible to human hepatoma cell lines as well as primary hepatocytes and have been successfully used as hepatocyte carriers in bioartificial liver devices (Damania et al., 2018). In this study, interpenetrating poly(*N*-isopropylacrylamide-co-gelatin) (pNG) cryogels were synthesized in 48 different compositions to yield scaffolds of different physicochemical properties. It was implemented that the interpenetration of biopolymer gelatin on synthetic poly(*N*-isopropylacrylamide) (pNIPAAm) modulates the microarchitecture, swelling properties, and compressive moduli of the scaffolds mainly by softening the cryogels. The cryogel scaffolds mimicking stiffness of normal, fibrotic, and cirrhotic liver were then utilized to study their effect on viability and functionality of human hepatocellular carcinoma cells. We believe that this study will provide a novel insight into the effect of interpenetrating polymer network on substrate properties and in turn the behavior of hepatic cells grown on these substrates.

2 | MATERIALS AND METHODS

2.1 | Materials

NIPAAm, *N*, *N*'-methylenebisacrylamide (MBAAm), and glutaraldehyde were purchased from Sisco Research Laboratories (Mumbai, India). Glycine and Tris were obtained from Merck (Mumbai, India). Dulbecco's modified Eagle's medium (DMEM) and antibiotic-antimycotic solution were procured from Hi-Media (Mumbai, India). Fetal bovine serum (FBS) and resazurin were obtained from Gibco-Thermo Fisher Scientific (MA, USA) and Loba Chemie (Mumbai, India), respectively. β -Nicotinamide adenine dinucleotide (β -NADH), sodium pyruvate, fluorescein diacetate (FDA), propidium iodide (PI), and dexamethasone were purchased from Sigma-Aldrich (MO). All other chemicals and reagents used were of analytical grade.

2.2 | Synthesis of pNG cryogels

The pNG cryogels were synthesized by a previously described method, only the concentrations of gel precursors were varied (Table S1; Sarkar & Kumar, 2016; Sarkar et al., 2017). The cryogels are represented as NxGy, where x and y correspond to initial concentrations (% wt/vol) of NIPAAm and gelatin, respectively. The monomer/polymer to crosslinker ratio was 2:1 for NIPAAm:MBAAm and 8:1 for gelatin:glutaraldehyde. After synthesis at -12°C for 16 h, the cryogels were washed with deionized water and the unreacted aldehyde groups of glutaraldehyde were blocked by immersing the cryogels in 0.5 M glycine in Tris buffer, pH 7.0 at 37°C for 1 h.

2.3 | Characterization of pNG cryogels

The cryogels were first dried under vacuum, cut into discs of 5 mm height and the dry weight was recorded. The dried cryogels were first subjected to Fourier-transform infrared (FTIR) spectroscopy analysis using JASCO FT/IR-6100. After sputter coating with platinum, the

cryogel disks were subjected to scanning electron microscope (SEM; TESCAN VEGA LSH) analysis. For swelling kinetics studies, the dried cryogel discs were submerged in PBS, pH 7.4 and at regular intervals of time, the weight was recorded. The water uptake capacity was calculated as follows:

$$W_u = \frac{M_t - M_d}{M_e} \times 100,$$

where W_u is percentage of water uptake capacity, M_t , M_d , and M_e represents weight of the swollen gel at time t , dry gel and swollen gel at equilibrium, respectively. The equilibrium degree of swelling was given by:

$$EDS = \frac{M_e}{M_d}.$$

The apparent porosity (AP) of the gels was calculated by the following formula:

$$\%AP = \frac{\text{Soaked weight} - \text{Dry weight}}{\text{Soaked weight} - \text{Suspended weight}} \times 100.$$

For protein adsorption studies, the dry cryogel discs were first treated with 70% ethanol for 30 min and then with absolute ethanol for 10 min. Thereafter, they were subjected to three washes of 15 min each with PBS, pH 7.4 and then exposed to UV for 20 min. The cryogels were then incubated in 10% FBS in PBS, pH 7.4 at 37°C and 5% CO₂ overnight. The protein adsorbed on the gel was measured by Bradford reagent by subtracting the protein left in the solution (after incubation with scaffold) from control (without scaffold). The mechanical strength of the swollen cryogels (5 mm height) was measured under uniaxial compression with 200 N load cell at a displacement of 0.5 mm/min using Bose® ElectroForce® 3200. The stress versus strain curve was plotted and compressive moduli were calculated from the slope in the range of 0–15% strain.

2.4 | Cell culture and seeding

The cryogel scaffolds were incubated in 70% ethanol for 30 min and then 100% ethanol for 10 min followed by three washes with PBS, pH 7.4 for 15 min each, and UV treatment for 20 min. The scaffolds were then equilibrated overnight in DMEM with 10% FBS and 1% antibiotic–antimycotic solution at 37°C and 5% CO₂. Human hepatocellular carcinoma cell lines, HepG2 and Huh-7 (obtained from National Center for Cell Sciences, Pune, India) were maintained in DMEM with 10% FBS and 1% antibiotic–antimycotic solution at 37°C and 5% CO₂. In the case of Huh-7, 1 mM of sodium pyruvate was also added to the culture medium. For seeding, live cells were counted using the trypan blue exclusion method in a hemocytometer and 2.5×10^4 cells were seeded on each 3D scaffold (2 mm height) and in each well of 2D 48-well tissue culture plates (2D control). The media was replaced every 48 h.

2.5 | Viability, lactate dehydrogenase release assay, and live–dead staining

After removing the spent media, the 3D scaffolds and wells of 2D control were washed with PBS, pH 7.4 and 250 µl resazurin (10 µg/ml in DMEM) was added. The fluorescent intensities were measured after 2 hr of incubation at 37°C and 5% CO₂ using Hidex Sense Microplate Reader at excitation/emission wavelength of 530/590 nm. Individual standard curves for 3D scaffolds and 2D control were constructed using the fluorescent intensities to estimate the cell number. The cytotoxicity of the cryogel scaffolds was determined by measuring the lactate dehydrogenase (LDH) released by the cells by a method previously described (Sarkar & Kumar, 2016). Live–dead staining of the cell-laden cryogel was accomplished using FDA and PI. FDA (0.1 mg/ml in acetone) and PI (2 µg/ml in deionized water) was mixed in 1:1 ratio, added to the scaffolds and observed under a fluorescent microscope (Leica DM 3000 LED).

2.6 | Albumin synthesis, cytochrome P450 3A4 activity and real-time polymerase chain reaction analysis

The albumin (ALB) released by the HepG2 in media was measured by enzyme-linked immunosorbent assay (ELISA) using Human Albumin ELISA Quantitation Set, Bethyl Laboratories, Inc. and cytochrome P450 3A4 (CYP3A4) activity was quantified by using P450-Glo™ CYP3A4 Assay System, Promega Corporation as per manufacturer's instruction. Hepatic gene expression of the HepG2 cells was analyzed by real-time polymerase chain reaction (PCR) for ALB and CYP3A4. Glyceraldehyde 3-phosphate dehydrogenase (GAPDH) was used as endogenous control. The primer sequences are depicted in Table S2. After synthesis of cDNA using Applied Biosystems cDNA synthesis kit, real-time PCR was carried out in Bio-Rad CFX™ Real-Time System in 10 µl reaction mixture with 1 µM primers and 5 µl Bio-Rad SYBR® Green Master Mix. The relative mRNA quantity was calculated by the comparative cycle threshold method and normalized by GAPDH values.

2.7 | Statistical analyses

All experiments were repeated three times and the results are represented as mean ± standard deviation. Furthermore, the data were analyzed by Student's *t* test.

3 | RESULTS AND DISCUSSION

3.1 | Synthesis and characterization of pNG cryogel scaffolds

The interpenetrating pNG cryogels were synthesized in 48 different compositions to yield scaffolds of different compressive moduli and are represented as NxGy where N and G represents NIPAAm and gelatin while x and y denotes the initial concentration (% wt/vol) of

NIPAAm and gelatin, respectively ($x = 0.5, 1, 2, 4, 6, 8, 10, 20$ and $y = 0, 0.5, 1, 1.5, 2, 2.5$). NIPAAm is a biocompatible material that mainly provided the structural integrity to the scaffold while gelatin provided the cell-adherent RGD motifs. All the gels exhibited stable structural integrity except N0.5G2.5 which showed a paste-like consistency and got powdered upon drying. Therefore, analyses other than FTIR were not possible in the case of N0.5G2.5.

3.1.1 | FTIR spectroscopy analysis

The incorporation of NIPAAm and gelatin in the scaffold was confirmed by FTIR analysis of Nx, NxG0.5, and NxG2.5 gels (Figure 1). All the gels, except N0.5G2.5, showed peaks at 1,155–1,175, 1,365–1,385, and 1,550–1,638 cm^{-1} corresponding to stretching vibration of isopropyl group, $-\text{CH}_3$ deformation vibration of isopropyl group, N–H bending, and amide $\text{C}=\text{O}$ bending of NIPAAm (Ren et al., 2015). N0.5G2.5 did not show any peak for NIPAAm signifying the NIPAAm was not incorporated in it, a reason why it did not exhibit proper structural integrity and formed a paste-like consistency. This

might be because the concentration of gelatin was much higher as compared with NIPAAm and it created a hindrance due to which NIPAAm was not able to polymerize. All the gels containing gelatin, except N0.5G0.5, showed peaks at 1,238 and 1,520 cm^{-1} corresponding to N–H bending of Amide III and in-plane N–H bending, C–N, and C–C stretching vibration of Amide II, respectively (Treesuppharat et al., 2017). N0.5G0.5 did not exhibit any peak for gelatin signifying that gelatin was not incorporated in it. This might be due to the very low concentration of NIPAAm that was unable to hold the gelatin network and led to washing off of the later. The concentration of gelatin was also lower to form an integrated framework. The gels N0.5Gy, N1Gy, and N2Gy showed the redshift of the peak of $\text{C}=\text{O}$ and C–N stretching vibration of Amide I at 1,647 cm^{-1} signifying the crosslinking of gelatin by glutaraldehyde. This redshift was not observed in gels with NIPAAm concentration of 4% wt/vol and above. This might be due to the high concentration of NIPAAm that created hindrance in crosslinking of gelatin and the polymer chains of gelatin were simply physically incorporated in the pNIPAAm network.

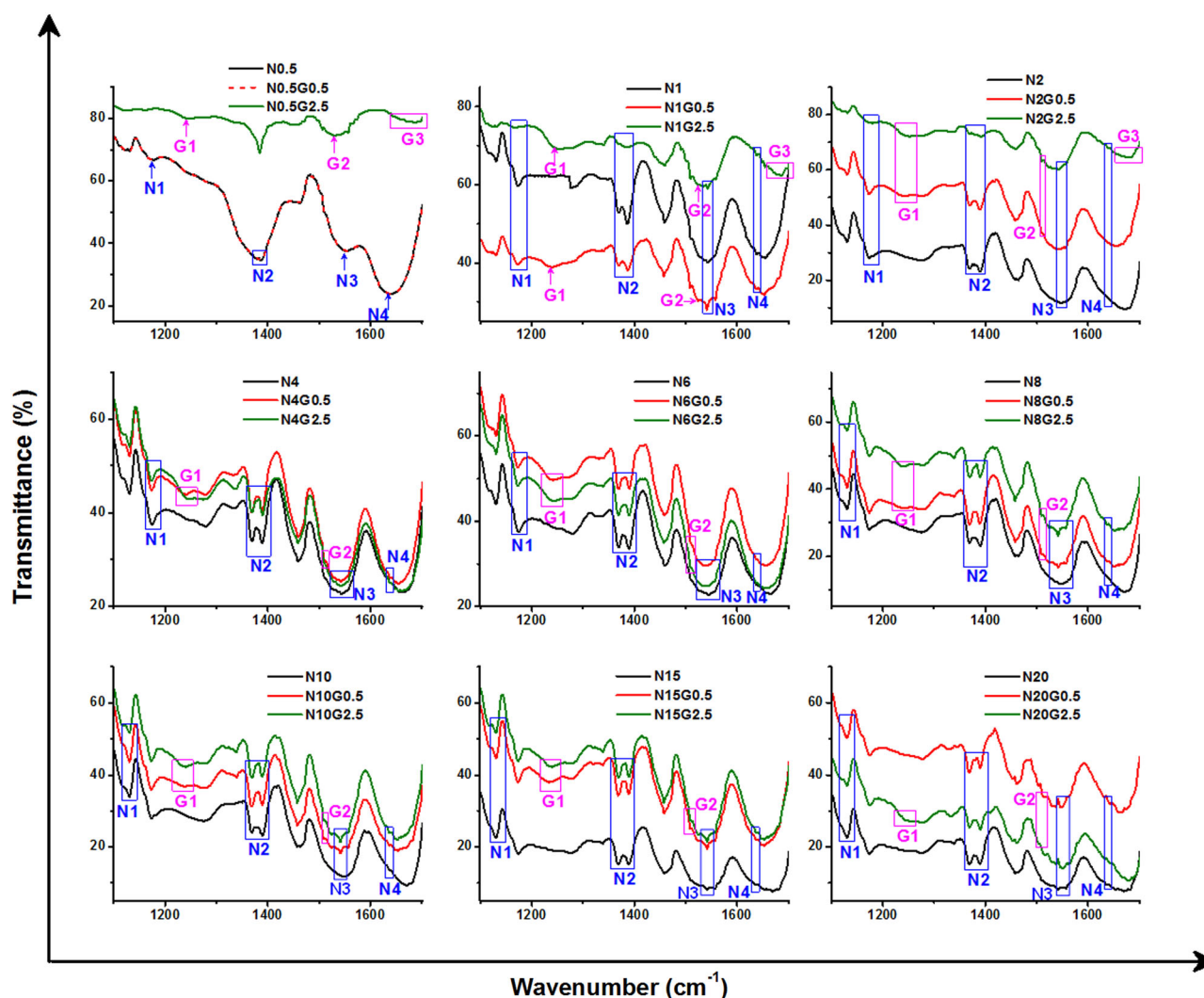


FIGURE 1 FTIR graphs of Nx, NxG0.5, and NxG2.5 pNG cryogel scaffolds; FTIR, Fourier-transform infrared; pNG, poly(*N*-isopropylacrylamide-co-gelatin) [Color figure can be viewed at wileyonlinelibrary.com]

3.1.2 | SEM analysis

To analyze the microstructure and pore morphology, SEM images of the cryogels were employed. The SEM images of the cryogel scaffolds reveal a porous structure with polymeric walls providing high surface area for cell attachment and accommodating pores that may allow the growth of cellular spheroids (Figure 2). In the case of N0.5Gy gels, improved pore architecture was observed upon an increase in gelatin concentration. The average pore size in the case of N0.5G0 and N0.5G0.5 was greater than 100 μm and around 70–100 μm in the case of N0.5G1 and N0.5G2 gels. The average pore size in the case of N1Gy–N6Gy was in the range of 70–100 μm with good structural integrity. Many closed pores were evident in the case of N8Gy gels that opened up with the increase in gelatin concentration. The effect was more pronounced in the case of the N10 group. The average pore size of these two was also less than 70 μm . N15 and N15G0.5 had no pores but the latter had cracks. Pores started appearing in the periphery of N15G1 gels. Small pores appeared in the case of N15G1.5 and the pore size increased in N15G2 and N15G2.5. N20G0–N20G1.5 gels had no pores but cracks. N20G2 had very small pores along with cracks. Only N20G2.5 had the proper porous structure of a cryogel. The average pore size of both N15 and N20 groups was less than those of N0.5–N10 gels. The cryogel microstructure indicates that a higher concentration of NIPAAm leads to reduced porosity and gelatin interpenetration on the pNIPAAm framework might render a porous structure; the effect is more evident when high concentration of NIPAAm is used.

3.1.3 | Swelling kinetics, porosity and protein adsorption studies

The swelling kinetics of the pNG cryogels showed high water uptake capacities (Figure 3a). Equilibrium water uptake capacity was attained within 20 s by all the gels except N15 and N15G1. This can be attributed to the high porosity and interconnected pores of the cryogels. In the case of N15, there were no pores and pores were present only in the periphery of N15G1 that led to their slow water uptake. However, in the case of N15G0.5 and N20G0–N20G1, the water uptake was fast due to the presence of cracks in the gels which was evident in the SEM images. High equilibrium water uptake capacity of approximately 90% was revealed in the case of N0.5–N8 cryogel groups. N10G0, N15G0, and N20G0 exhibited a drop in water uptake capacity to approximately 80%, 70%, and 60%, respectively. However, with an increase in gelatin concentration in these groups, the uptake capacity of the gels got better that might be due to improved porous structure (Figure 3b). The equilibrium degree of swelling decreased with an increasing concentration of NIPAAm from N0.5–N20. The increase in the polymer concentration restricted the swelling of the cryogel. Among each group, the degree of swelling increased with increasing concentration of gelatin from G0–G2.5 (Figure 3c). The results of equilibrium water uptake capacity and degree of swelling are in accordance with swelling kinetics of the gels indicating that gelatin interpenetration modulates

the microstructure of the gels and improves its porosity. Similar to swelling properties, the AP of the gels decreased with an increase in NIPAAm concentration from N0.5–N20 owing to the increased compactness of the gels. In the case of N10–N20, the effect of gelatin interpenetration was evident in terms of increasing AP from G0–G2.5 (Figure 4a). Both swelling behavior and porosity of the scaffolds are suggestive of better exchange of nutrients and gases and less accumulation of wastes in the cryogels. Assessment of protein adsorbed on the scaffolds is another important parameter of the scaffold as it mediates the interaction of cells to the surface of the scaffold. The protein adsorption on the gel also increased with increasing NIPAAm concentration owing to increased surface area (Figure 4b). However, no effect of gelatin interpenetration was observed because there was a negligible increase in the surface area upon gelatin incorporation.

3.1.4 | Analysis of compressive moduli

Furthermore, the compressive moduli at 15% strain of the cryogels were analyzed and it was observed that the stiffness of the gels increased with an increase in the concentration of NIPAAm confirming that it enhances the structural integrity of the gel. This might be because the increase in NIPAAm leads to the formation of the packed network leading to the formation of a rigid gel. The lowest modulus was that of N0.5 (0.371 kPa) and the highest was that of N20 (1,404.5 kPa; Figure 5a,b). However, the increase in the concentration of gelatin led to a reduction in compressive moduli indicating that gelatin interpenetration in pNIPAAm network softens the cryogel. The only exception was the N0.5 group in which the moduli increased with an increase in gelatin concentration corroborating with SEM analyses that showed better gel architecture upon an increase in gelatin concentration in the N0.5 group. This confirms that 0.5% wt/vol is a very low concentration of NIPAAm to achieve optimum structural integrity. Interestingly, as the concentration of NIPAAm increases, the effect of gelatin interpenetration in reducing the compressive moduli expanded. For example, the range of moduli for N1G0–N1G2.5 is 4.42–1.93 kPa but that of N10G0–N10G2.5 is 380.17–59.18 kPa (Figure 5c). The individual values of the compressive moduli of each cryogel scaffold are represented in Table S3. As evident from the SEM images, there were cracks in N15G0–G0.5 and N20G0–G1.5 gels; therefore, measurement of compressive moduli was not possible for these cryogels. It can be concluded from the compressive moduli studies that the stiffness of the pNG cryogel scaffolds can be tailored by modulating the concentration of the gel precursors and optimum moduli can be obtained to mimic the ECM stiffness of various tissues. Overall, the physicochemical characterization of the pNG cryogels indicates that apart from imparting cell-adherent motifs, gelatin interpenetration in pNIPAAm network modulates the properties of the cryogel. The interpenetration leads to gain in porous structure and hence improving the swelling properties and porosity of the gel. Most interestingly, it softens the stiffness of the cryogel by decreasing the compressive moduli. A probable explanation to this modulation could be that increasing gelatin concentration hinders the crosslinking of the pNIPAAm network leading to decreased

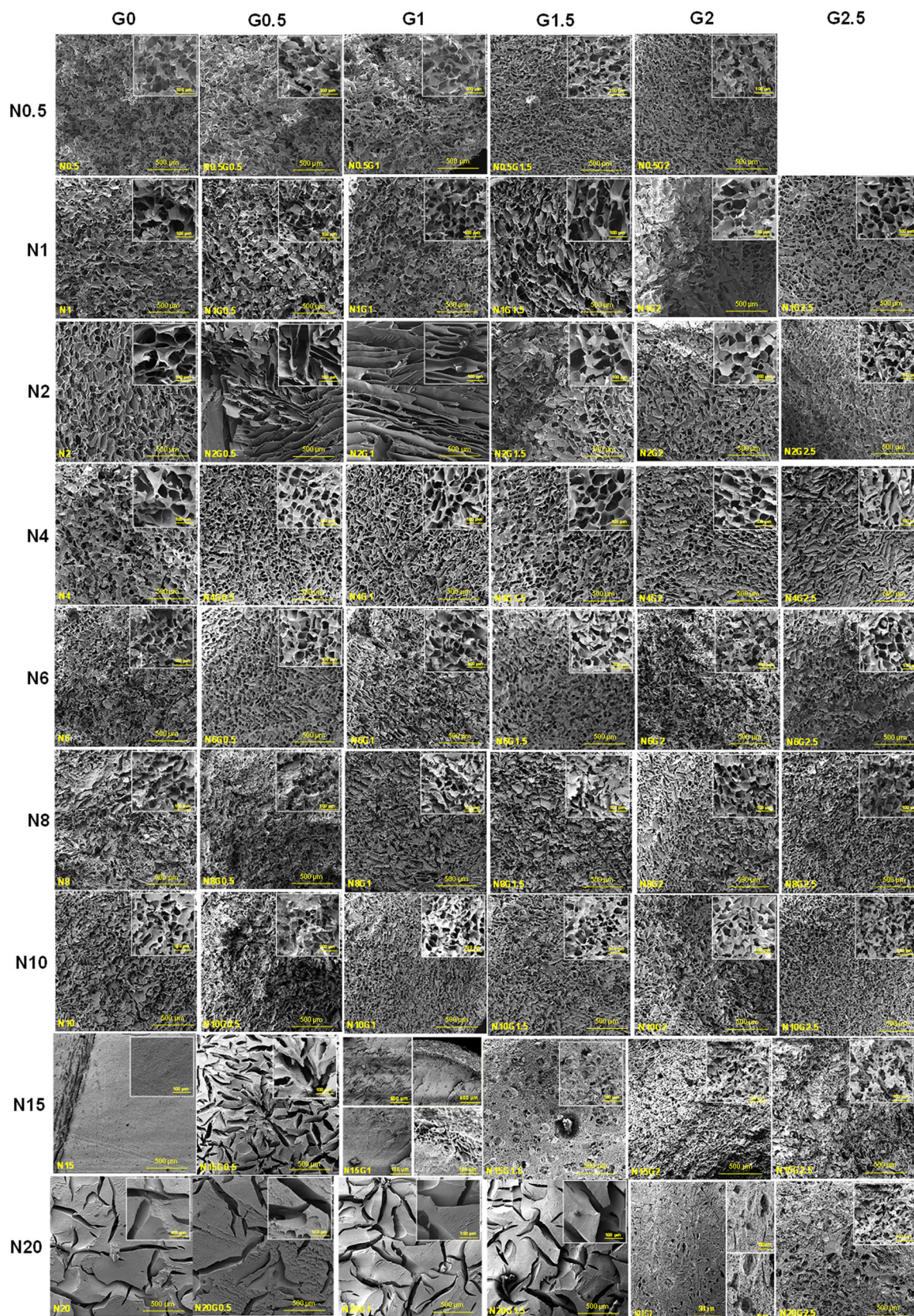


FIGURE 2 SEM images of NxG0–NxG2.5 pNG cryogel scaffolds. pNG, poly(N-isopropylacrylamide-co-gelatin); SEM, scanning electron microscope [Color figure can be viewed at wileyonlinelibrary.com]

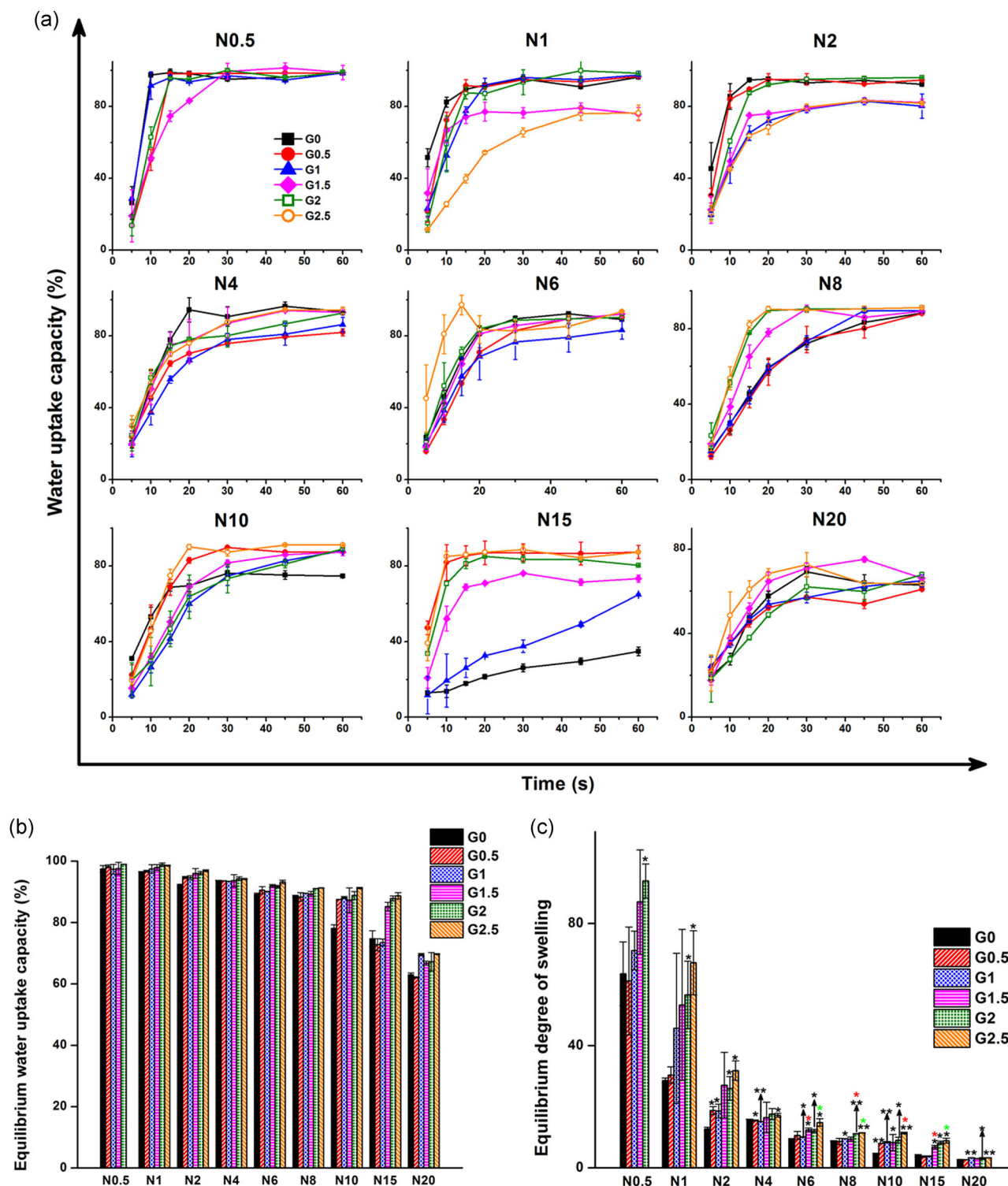


FIGURE 3 Swelling properties of NxG0-NxG2.5 pNG cryogel scaffolds. (a) Swelling kinetics. (b) Equilibrium water uptake properties. (c) Equilibrium degree of swelling. * $p \leq .05$; ** $p \leq .01$. Black asterisk indicates p value in comparison to NxG0 cryogel while red and green asterisk indicate p value in comparison to the preceding gel. pNG, poly(*N*-isopropylacrylamide-co-gelatin) [Color figure can be viewed at wileyonlinelibrary.com]

rigidity and improved porosity (Jaiswal et al., 2010). In case of lower concentration of NIPAAm, the effect of gelatin gets saturated; a reason why the effect becomes more pronounced at higher concentrations of NIPAAm.

3.2 | Cell material interaction

Cytocompatibility of the gels was examined by culturing human hepatocellular carcinoma cell line, HepG2 on Nx, NxG0.5, and

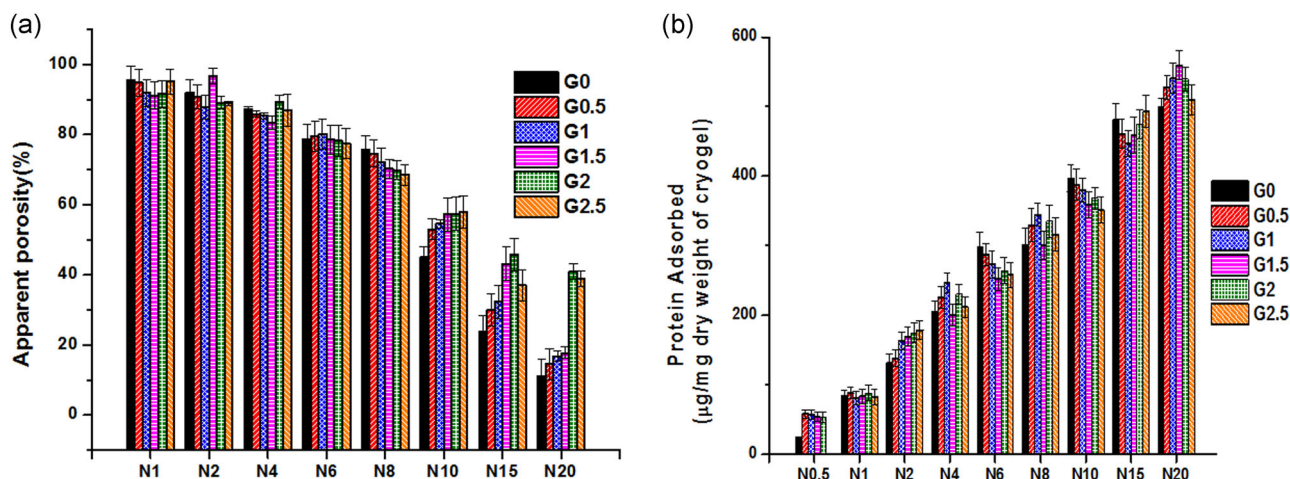


FIGURE 4 (a) Apparent porosity and (b) protein adsorption of the pNG cryogel scaffolds. pNG, poly(*N*-isopropylacrylamide-co-gelatin) [Color figure can be viewed at wileyonlinelibrary.com]

NxG2.5 gels. The viability of the cells was analyzed by resazurin reduction and LDH release after 24 h of seeding on the scaffolds. Furthermore, the microscopic analysis was accomplished by the live/dead staining of the cell-laden scaffolds by FDA and PI. Resazurin is

reduced to resorufin by the mitochondrial dehydrogenases of viable cells that can be measured spectrofluorimetrically (Vega-avila & Pugsley, 2011). The fluorescent intensities of the reduced resazurin exhibit that the viability of the cells was low in the case of the N0.5

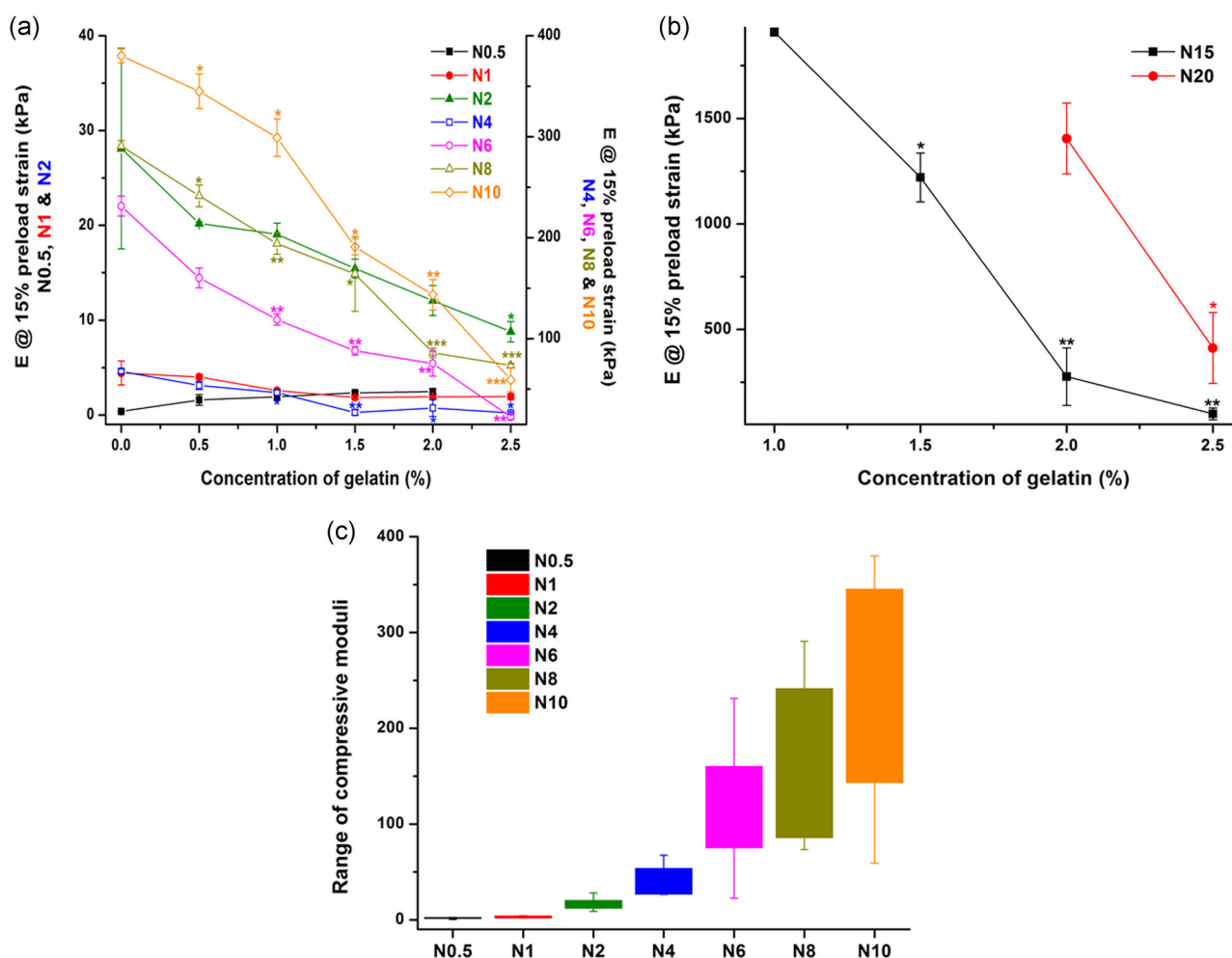


FIGURE 5 Compressive moduli of the pNG cryogel scaffolds. (a) N0.5Gy to N10Gy. (b) N15Gy and N20Gy. * $p \leq .05$ and ** $p \leq .01$. (c) Box and whisker plot showing a range of moduli of the cryogels. pNG, poly(*N*-isopropylacrylamide-co-gelatin) [Color figure can be viewed at wileyonlinelibrary.com]

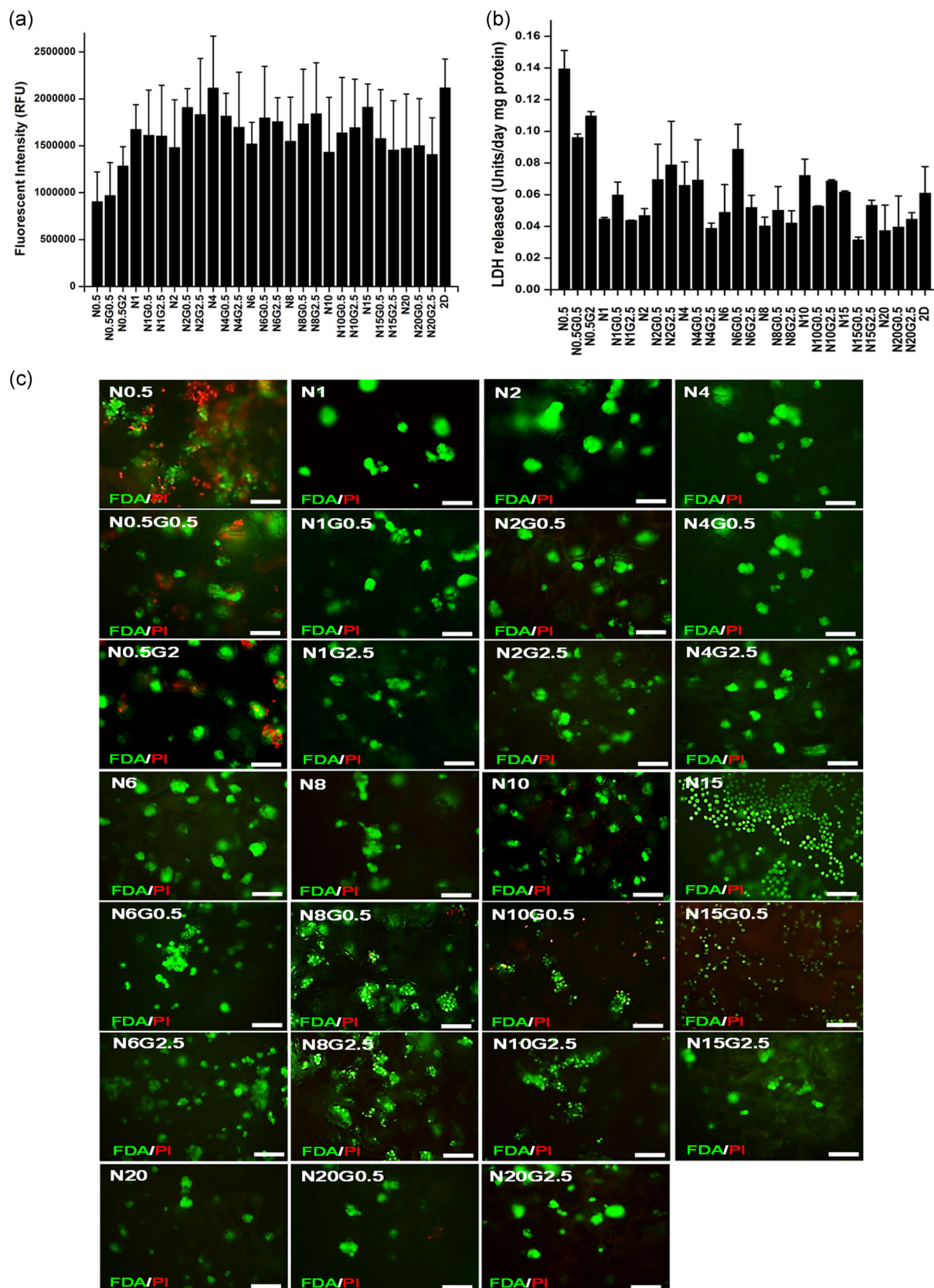


FIGURE 6 Viability of HepG2 cells on pNG cryogel scaffolds. (a) Resazurin reduction. (b) Lactate dehydrogenase release. (c) Live/dead staining. In each well 2.5×10^4 cells were seeded and cultured for 24 h. pNG, poly(*N*-isopropylacrylamide-co-gelatin) [Color figure can be viewed at wileyonlinelibrary.com]

group. This might be due to low polymer concentration leading to less adherent area for the cells. There was no significant difference in the viability of the cells in the rest of the cryogel scaffolds (Figure 6a). A similar trend was observed in the case of LDH release assay of the spent media and live-dead staining of the cell-laden scaffolds. LDH is released in the spent media due to membrane leakage of dying cells (Iguchi et al., 1993). LDH release was higher in the case of the N0.5 group indicating membrane disintegration and cell death (Figure 6b). FDA was used to mark the live cells; FDA is taken up by the cells, cleaved to fluorescein by esterases which is impermeable to the cell membranes and hence stays within the cells. PI, on the other hand, is impermeable to the cell membranes and passes through the disintegrated membrane of dying cells, thereby probing dead cells (Fleddermann et al., 2019). Similar to resazurin reduction and LDH release, more PI stained (red), that is, dead cells were visible in the N0.5 group as compared to other scaffolds wherein FDA stained (green), that is, viable cells are visible (Figure 6c).

In addition, the growth rate of HepG2 and another human hepatocarcinoma cell line, Huh-7 on the cryogel scaffolds was also analyzed (Figure S1). Both the cells showed a high rate of proliferation in 2D culture as compared to 3D culture systems. But, there was a drop in the proliferation of both the cells after 96 h of culture in case of 2D indicating cell death which might be due to contact inhibition. In the case of 3D culture, both HepG2 and Huh-7 cells exhibited an increase in cell viability till 120 h; however, the fluorescent intensity as measured by resazurin reduced by the cells was much lower as compared with 2D culture. This indicates that the cells remain viable in 3D cryogel scaffolds but their proliferation is arrested. The doubling time of the cells was found to be approximately 48 h for HepG2 and 24 h for Huh-7 as estimated by relative fluorescent intensity values. Altogether, the cytocompatibility studies indicate that the gels are cell-adherent in nature and are suitable for the culture of hepatic cells.

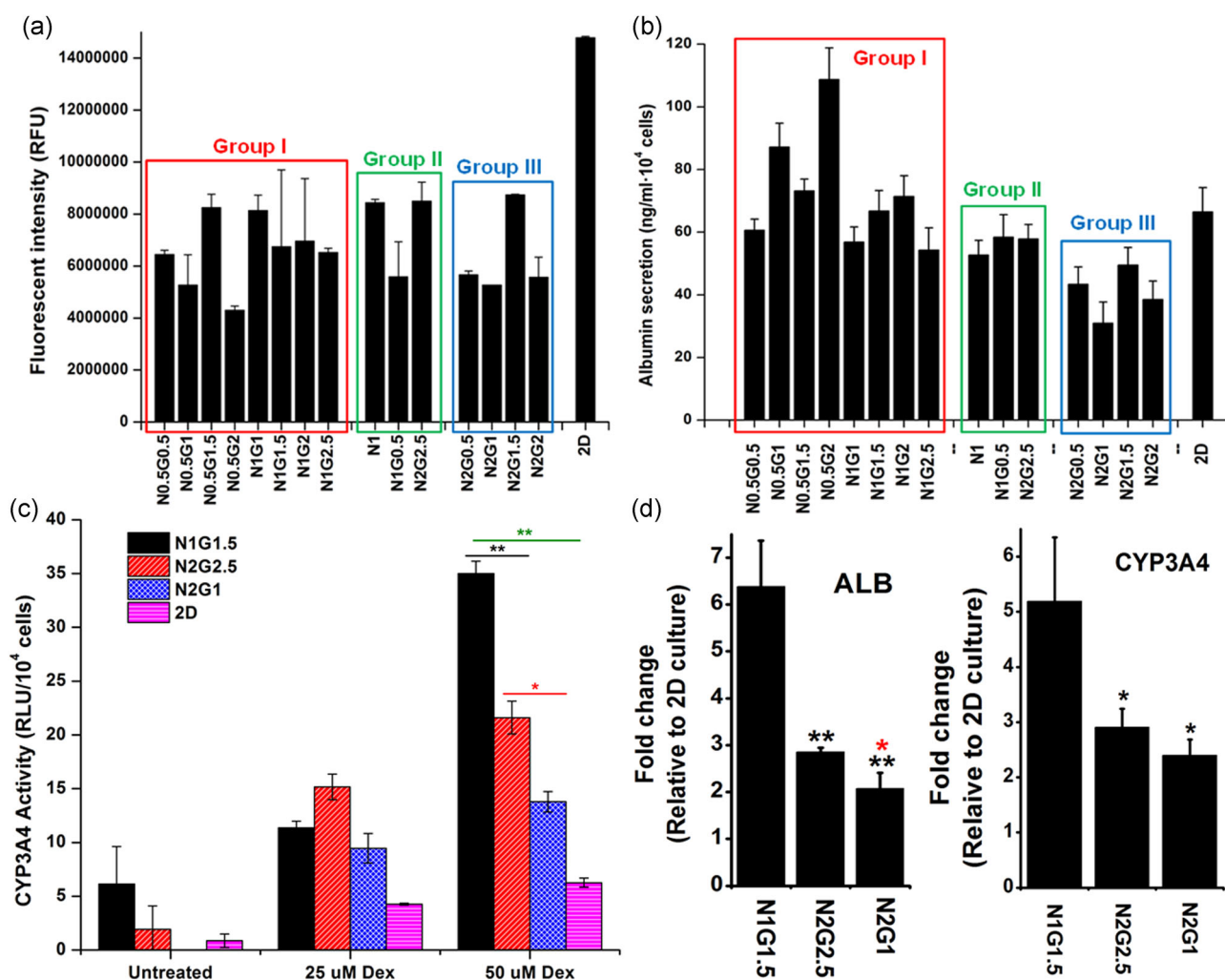


FIGURE 7 Liver-specific functionality of HepG2 cells on pNG cryogel scaffolds. (a) Resazurin reduction. (b) Albumin secretion. (c) CYP3A4 activity. (d) Real-time PCR analysis of expression of albumin (ALB) and CYP3A4 genes (Black and the red asterisk indicate p value in comparison to N1G1.5 and N2G2.5, respectively). HepG2 cells (2.5×10^4) were seeded on pNG cryogel scaffolds of different stiffness and analyses were done on Day 5 of culture (Dex, dexamethasone). * $p \leq .05$; ** $p \leq .01$. PCR, polymerase chain reaction; pNG, poly(*N*-isopropylacrylamide-co-gelatin) [Color figure can be viewed at wileyonlinelibrary.com]

3.3 | Liver-specific functionality of HepG2 and Huh7 cells on pNG cryogel scaffolds

On the basis of the physicochemical properties of the synthesized cryogels and their interaction with HepG2 cells, they were divided into three groups corresponding to three major stages of diseased liver namely, normal, fibrotic, and cirrhotic. These three diseased stages of the liver are characterized by the compressive moduli of the ECM; with the progression of the disease from normal to cirrhotic, the matrix gets stiffer. Therefore, the cryogel scaffolds were distributed in Group I (N0.5G0.5, N0.5G1, N0.5G1.5, N0.5G2, N1G1, N1G1.5, N1G2, N1G2.5), Group II (N1, N1G0.5, N2G2.5), and Group III (N2G0.5, N2G1, N2G1.5, N2G2) with compressive moduli of 1.5–2.5, 4–9, and 10–20 kPa correlating to normal, fibrotic, and cirrhotic liver, respectively (Mueller & Sandrin, 2010). Corroborating with the previous results of viability, the resazurin reduction by HepG2 seeded on scaffolds in the above three groups did not show any significant difference by Day 5 of culture. The resazurin reduction by HepG2 was much higher in the case of 2D control

as compared with 3D scaffolds indicating high proliferation in 2D monolayer and arrest in proliferation in a 3D culture which might be due to cell cycle arrest (Figure 7a; Li et al., 2008). However, the liver-specific functionalities, that is, ALB synthesis and CYP3A4 metabolic activity of the HepG2 cells cultured on 3D cryogel scaffolds showed a different trend. There was significant difference in ALB production between cells seeded in Group I (normal) and Group III (cirrhotic) cryogels with higher ALB secretion by cells seeded in scaffolds of Group I, that is, normal liver stiffness (Figure 7b). However, there was no difference in ALB secretion by the cells seeded in the same group of scaffolds. Based on this, the cryogel scaffolds were further funneled down to three scaffolds namely, N1G1.5 (1.85 kPa), N2G2.5 (8.78 kPa), and N2G1 (19.06 kPa) representing Group I (normal), Group II (fibrotic), and Group III (cirrhotic) liver ECM, respectively. The basal activity of CYP3A4 of HepG2 on these scaffolds showed no significant difference; only, no basal CYP3A4 activity was detected in N2G1 (cirrhotic) scaffold. After induction with 25 μ M dexamethasone, better activity was evident in all the groups but there was no significant difference in activities

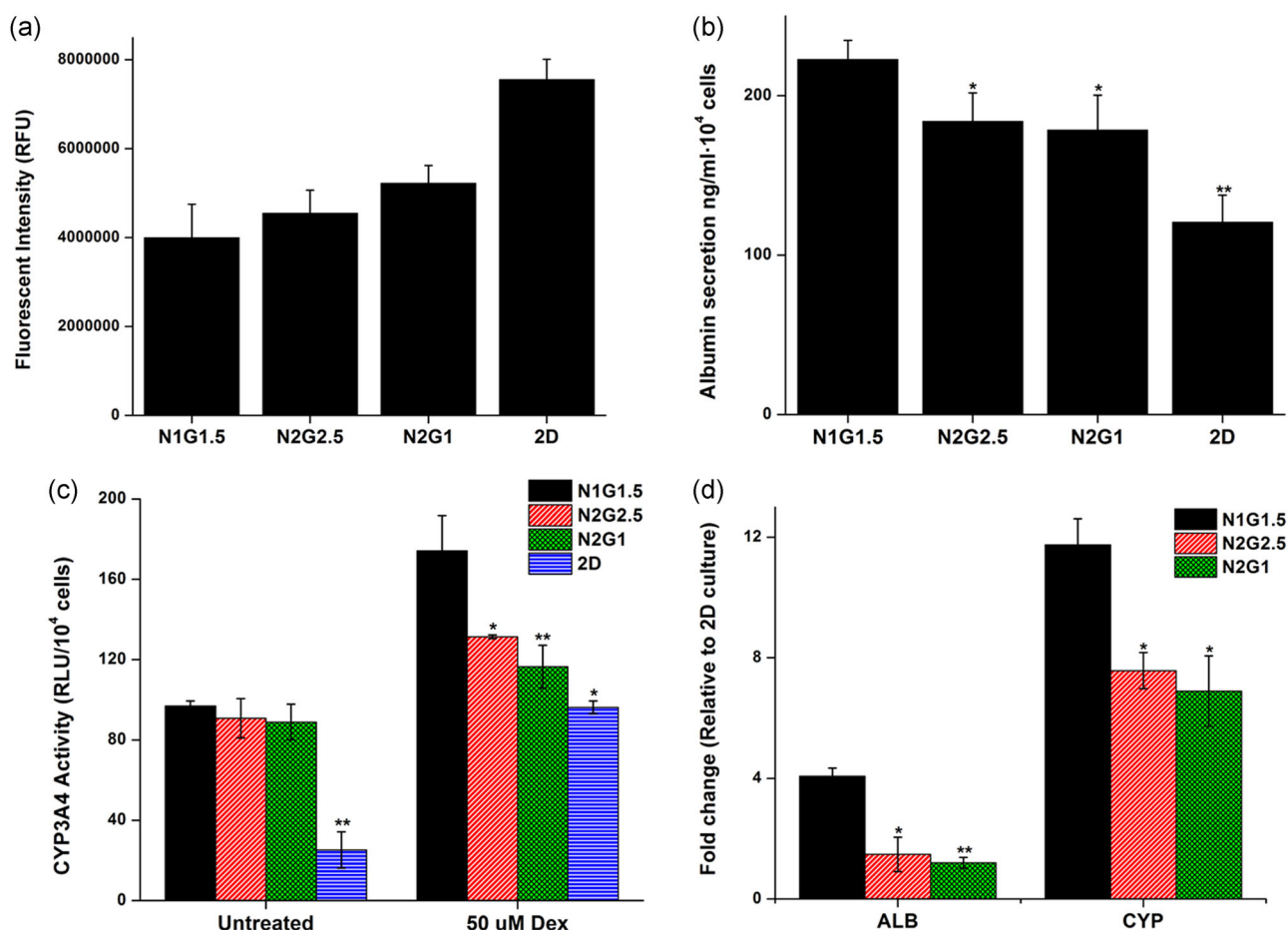


FIGURE 8 Liver-specific functionality of Huh-7 cells on pNG cryogel scaffolds. (a) Resazurin reduction. (b) Albumin secretion. (c) CYP3A4 activity. (d) Real-time PCR analysis of expression of albumin (ALB) and CYP3A4 genes (asterisk indicates p value in comparison to N1G1.5). Huh-7 cells (2.5×10^4) were seeded on pNG cryogel scaffolds of different stiffness and analyses were done on Day 5 of culture (Dex, dexamethasone). * $p \leq .05$; ** $p \leq .01$. PCR, polymerase chain reaction; pNG, poly(N-isopropylacrylamide-co-gelatin) [Color figure can be viewed at wileyonlinelibrary.com]

between the groups, except the activity of 2D culture was significantly lower as compared with others. However, in the case of induction of CYP3A4 activity with 50 μ M dexamethasone, the highest activity was exhibited by HepG2 cells seeded on N1G1.5 (normal) followed by N2G2.5 (fibrotic) and N2G1 (cirrhotic) cryogel scaffolds. Also, the activity of 2D control was much lower as compared with the 3D scaffolds (Figure 7c). The real-time PCR analysis also exhibited a similar trend. Figure 7d shows fold change in expression of ALB and CYP3A4 genes in comparison to 2D culture. It was observed that the relative expression of both these genes reduced significantly as the stiffness of the scaffold increased. The expression of ALB by cells cultured on N1G1.5, N2G2.5, and N2G1 was approximately 6.4, 2.8, and 2.1 times higher than 2D control, respectively. Similar to CYP3A4 activity, the highest expression of the CYP3A4 gene was revealed by cells cultured on N1G1.5 (~5.2 folds) followed by N2G2.5 (~2.9 folds) and N2G1 (~2.4 folds) cryogel scaffolds as compared with 2D control.

To, reinforce the findings, the study was repeated with another human hepatocarcinoma cell line, Huh-7. No significant difference in the viability of Huh-7 was observed when they were grown on N1G1.5 (normal), N2G2.5 (fibrotic), and N2G1 (cirrhotic) cryogel scaffolds by Day 5 of culture (Figure 8a). The hepatic functionality exhibited a similar trend like that of HepG2 cells. Both ALB secretion and CYP3A4 activity (after induction with 50 μ M dexamethasone) were significantly higher when Huh-7 cells were grown on N1G1.5 (normal) cryogel scaffolds (Figure 8b,c). This was also reflected in terms of gene expression wherein expression of ALB and CYP3A4 genes were decreased with an increase in scaffold stiffness. As compared with 2D culture, there was approximately 4.1, 1.5, and 1.2 folds increase in ALB and approximately 11.7, 7.6, and 6.9 folds increase in CYP3A4 gene expression when Huh-7 cells were cultured on N1G1.5 (normal), N2G2.5 (fibrotic), and N2G1 (cirrhotic) cryogel scaffolds, respectively (Figure 8d). The analyses of HepG2 and Huh-7 cells seeded on different scaffolds mimicking the stiffness of different stages of diseased liver reveal that to obtain optimum liver-specific functionalities, the cells should be cultured on scaffolds mimicking the stiffness of ECM of the normal liver. Increasing the stiffness to fibrotic and cirrhotic ECM leads to a decrease in functionality.

4 | CONCLUSION

In conclusion, gelatin interpenetration in the pNIPAAm network modulates the physicochemical properties of the pNG cryogels that in turn affect the biological activity of the cells cultured on these scaffolds. Specifically, it reduces the stiffness of pNG cryogel scaffolds which can be utilized to mimic hepatic ECM. Both HepG2 and Huh-7 cells exhibit better hepatic functionalities when cultured on pNG cryogel with a stiffness similar to normal liver. On the basis of the stiffness of the cryogels, differentiation of stem cells could be guided to a hepatic lineage which can be utilized for pharmaco-

toxicological analysis of probable drug compounds. The compressive moduli of the cryogel can be modulated to mimic the ECM stiffness of various disease stages of different tissues and thus can be utilized for disease modeling.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

ORCID

Joyita Sarkar  <http://orcid.org/0000-0003-3323-2759>

Ashok Kumar  <http://orcid.org/0000-0002-4910-9440>

Suresh W. Gosavi  <http://orcid.org/0000-0001-7540-0664>

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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